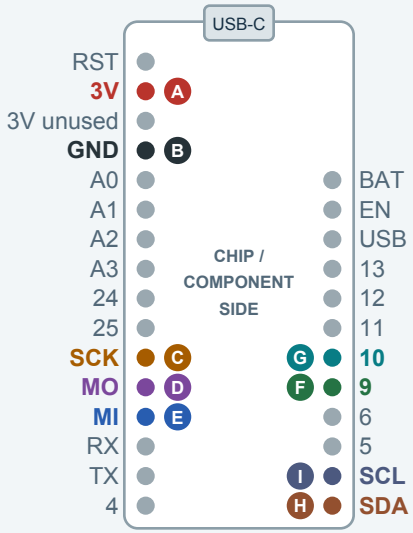
# blip / soldering map

Feather RP2040 to 2.4-inch TFT FeatherWing V2 | 9 direct wires | Keep the existing Feather headers

**POWER OFF: unplug USB and any battery. Follow pin labels; the drawings below are not to scale.**

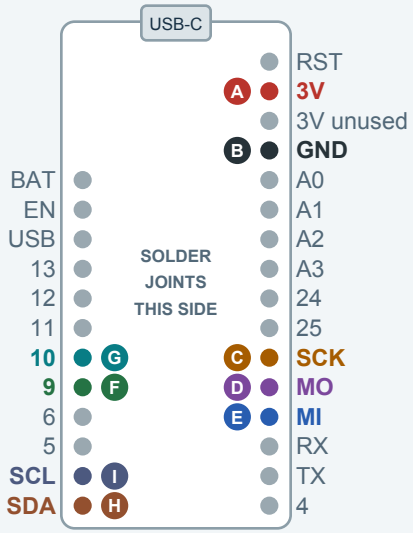
### FEATHER: component face

USB-C at TOP; use this to identify labels



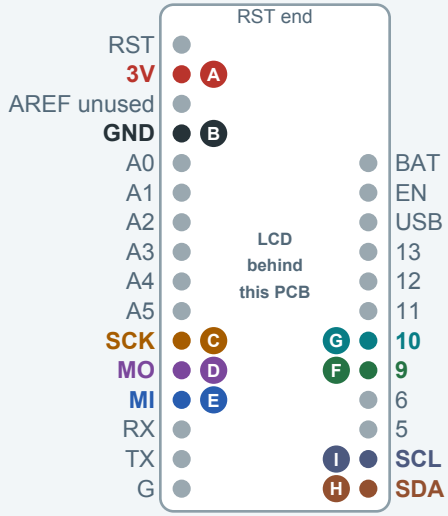

### FEATHER: solder-joint face

Flip left-to-right; keep USB-C at TOP




### SCREEN: header / pad map

Connector side facing you; RST end at TOP




**IMPORTANT: Feather A (3V) goes to screen A (3V). The next screen position is AREF - leave it unused.**

| Wire | Feather pin | Screen pad / socket | What it does |
|---|---|---|---|
| A | **3V / 3.3V** | **3V / 3.3V** | 3.3 V power - use the second position from RST |
| B | **GND** | **GND** | Ground / power return |
| C | **SCK** | **SCK / SCLK** | SPI clock: timing for screen commands and pixels |
| D | **MO / MOSI** | **MO / MOSI** | SPI data from Feather to screen |
| E | **MI / MISO** | **MI / MISO** | SPI return data: LCD state readback (new ninth wire) |
| F | **9 / D9** | **9 / D9 (TFTCS)** | Screen chip select |
| G | **10 / D10** | **10 / D10 (TFTDC)** | Selects command bytes versus pixel/data bytes |
| H | **SDA** | **SDA** | I2C data: touchscreen on the screen board |
| I | **SCL** | **SCL** | I2C clock: touchscreen on the screen board |

Letters/colors are wire IDs, not pin numbers. Gray pins stay unconnected. Knob: keep its QT cable plugged into the Feather.
Solder to the existing Feather pin joints; inspect for bridges. Screen square pads and matching sockets share the same signals.
Sources: Adafruit Feather RP2040 PCB + TFT FeatherWing V2 PCB / pinout guides. Layout is a reference; verify board labels.